

☒ Futu ML 機器學習回測報告

報告日期: 2026年07月05日 11:50 HKT
數據源: Futu OpenD 日K (前復權 QFQ, 5年)
模型: SVM-rbf 為主 + RandomForest 特徵重要性 | 41 技術因子
驗證: 10折 Walk-Forward (TimeSeriesSplit) | 起始資金 \$100,000 | 50% 倉位
基本面: PE/PB/ROE/ProfitMargin pre-filter (PE>40 或負數跳過, 除非高增長)
標的: 11 隻美港股 | 通過篩選: 9 | 被篩走: 2

☒ 基本面篩選未過之股票

股票	代碼	原因	基本面評分
3690美團	HK.03690	ROE=-458.84% 負數	25.0
3317訊策	HK.03317	ROE=-484.85% 負數	12.0

☒ 綜合排名 — 通過篩選之股票

排名	股票	類型	準確率	總回報	MaxDD	Sharpe	WinRate	交易	ProfitF	基本面
1	0388港交所	港股	57.34%	+37.73%	3.79%	4.25	61.1%	108	2.11	67.0
2	9988阿里	港股	55.14%	+28.14%	5.26%	3.87	61.4%	57	2.03	85.0
3	0700騰訊	港股	54.22%	+19.06%	5.97%	3.15	57.6%	85	1.71	—
4	7709湛江	港股	60.00%	+153.15%	38.14%	3.08	59.8%	97	1.67	50.0
5	MU	美股	52.43%	+244.60%	16.25%	1.86	53.7%	598	1.39	67.0
6	NVDA	美股	54.14%	+201.68%	19.35%	1.72	55.9%	732	1.34	—
7	GOOGL	美股	54.14%	+70.84%	15.97%	0.98	54.1%	929	1.18	57.0
8	META	美股	53.24%	+53.34%	19.95%	0.98	54.9%	576	1.22	—
9	MSFT	美股	54.86%	+36.95%	13.10%	0.86	54.8%	728	1.16	67.0

☒ 逐股詳細分析 + Equity Curve

0388港交所 (HK.00388) [港股]

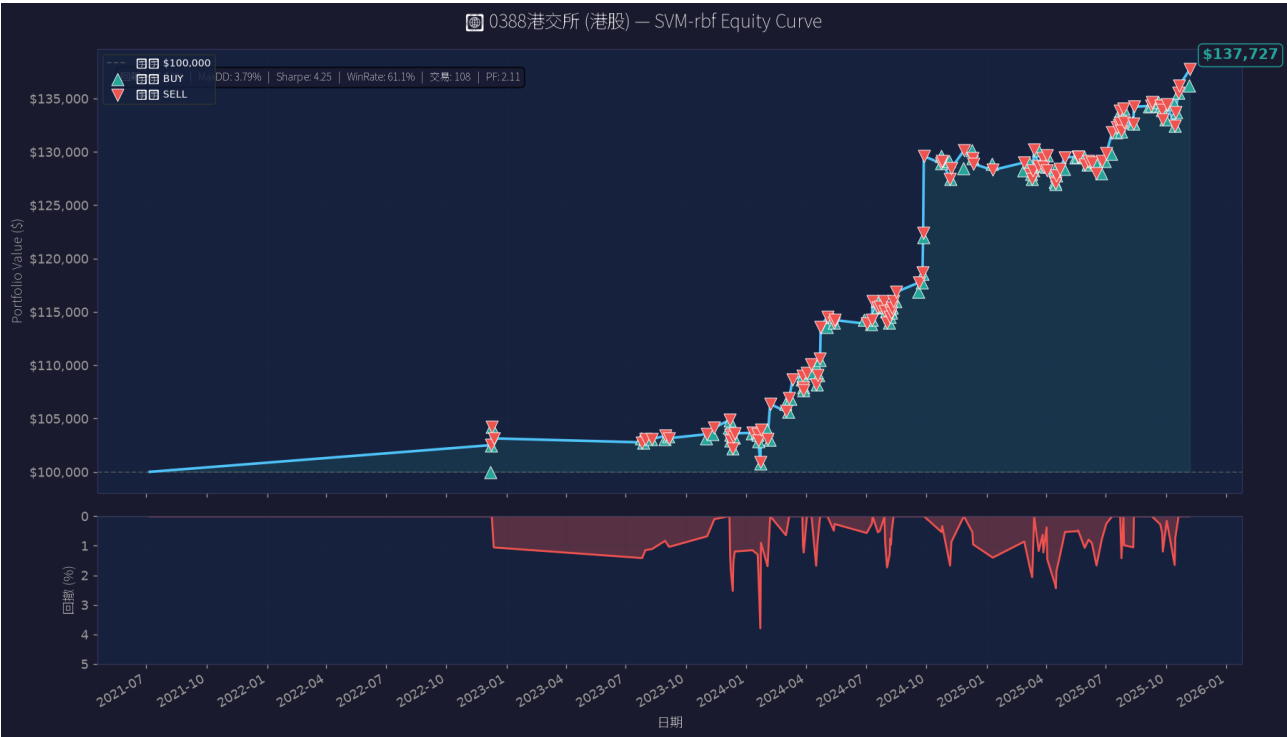
☒ 2021-08-11 ~ 2026-07-02 | 1199 bars | 41因子 | 基本面評分: 67.0

準確率	AccStd	Baseline	總回報	MaxDD	Sharpe	Sortino	Calmar	WinRate	ProfitF	AvgWin	AvgLoss	交易數
57.34%	4.11%	56.70%	+37.73%	3.79%	4.25	10.77	9.94	61.1%	2.11	1.89%	-1.41%	108

Fold 準確率: 58.7%, 60.6%, 61.5%, 56.9%, 61.5%, 59.6%, 55.0%, 47.7%, 53.2%, 58.7% | 參數: {"C": "0.1", "gamma": "scale", "kernel": "rbf", "max_iter": "5000", "probability": "True"}

基本面: pe_ttm=25.201 | pb=8.62 | roe=9.158149 | profit_margin=0.76106971 | revenue_growth_yoy=0.70435866 | market_cap=475438835625.0 | log_market_cap=26.88750407895462

☒ 特徵重要性 (RF): upper_shadow:0.044 | vol_ratio_5:0.036 | vol_ma5:0.035 | vol_ratio_20:0.034 | volatility_5d:0.033 | body_pct:0.029 | lower_shadow:0.028 | close_ma10:0.028



9988阿里 (HK.09988) [港股]

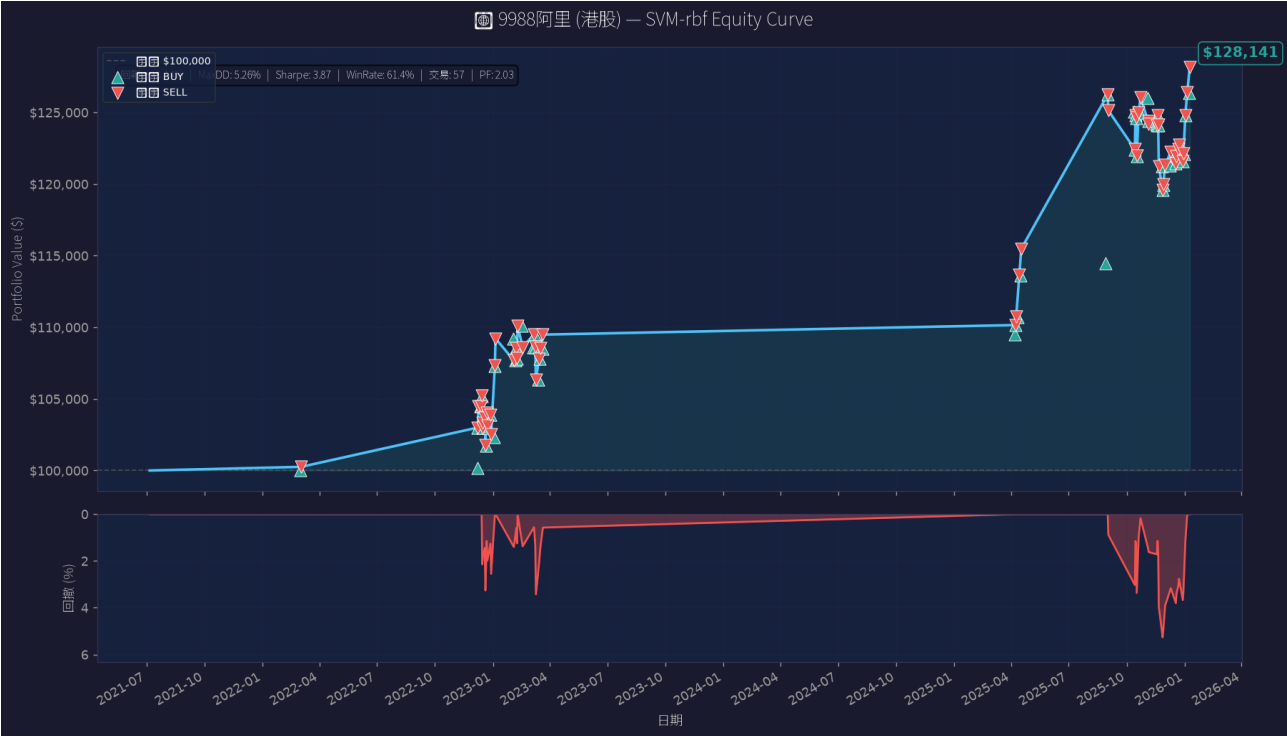
☒ 2021-08-11 ~ 2026-07-02 | 1199 bars | 41因子 | 基本面評分: 85.0

準確率	AccStd	Baseline	總回報	MaxDD	Sharpe	Sortino	Calmar	WinRate	ProfitF	AvgWin	AvgLoss	交易數
55.14%	3.66%	54.59%	+28.14%	5.26%	3.87	10.24	5.35	61.4%	2.03	2.91%	-2.39%	57

Fold 準確率: 56.9%, 59.6%, 56.0%, 54.1%, 56.0%, 52.3%, 50.5%, 50.5%, 53.2%, 62.4% | 參數: {"C": "10", "gamma": "scale", "kernel": "rbf", "max_iter": "5000", "probability": "True"}

基本面: pe_ttm=15.068 | pb=1.504 | roe=10.220694 | profit_margin=0.09976555 | revenue_growth_yoy=0.16834753 | market_cap=1807313929652.6 | log_market_cap=28.22286284219625

☒ 特徵重要性 (RF): vol_ma5:0.034 | obv_ma5:0.033 | close_ma5:0.032 | obv:0.031 | vol_ratio_5:0.030 | range_pct:0.030 | ma5:0.029 | vol_ratio_20:0.028



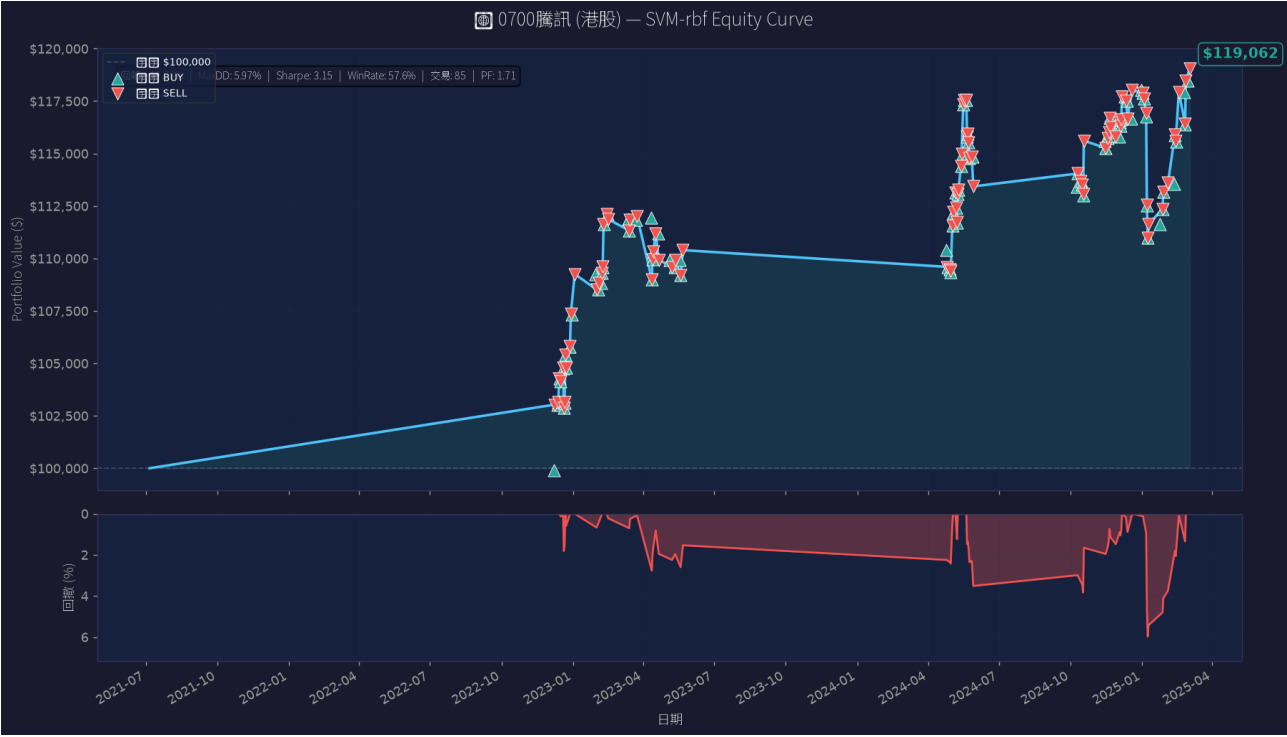
0700騰訊 (HK.00700) [港股]

☑ 2021-08-11 ~ 2026-07-02 | 1199 bars | 41因子 | 基本面評分: —

準確率	AccStd	Baseline	總回報	MaxDD	Sharpe	Sortino	Calmar	WinRate	ProfitF	AvgWin	AvgLoss	交易數
54.22%	3.90%	54.31%	+19.06%	5.97%	3.15	4.41	3.19	57.6%	1.71	1.76%	-1.44%	85

Fold 準確率: 56.0%, 56.0%, 57.8%, 56.0%, 52.3%, 51.4%, 55.0%, 45.9%, 51.4%, 60.6% | 參數: {"C": "0.1", "gamma": "scale", "kernel": "rbf", "max_iter": "5000", "probability": "True"}

☑ 特徵重要性 (RF): ret_10d:0.041 | range_pct:0.039 | rsi_7:0.037 | stoch_k:0.034 | vol_ratio_5:0.031 | close_ma10:0.030 | close_ma5:0.030 | volatility_5d:0.030



7709湛江 (HK.07709) [港股]

☑ 2025-12-29 ~ 2026-07-02 | 123 bars | 41因子 | 基本面評分: 50.0

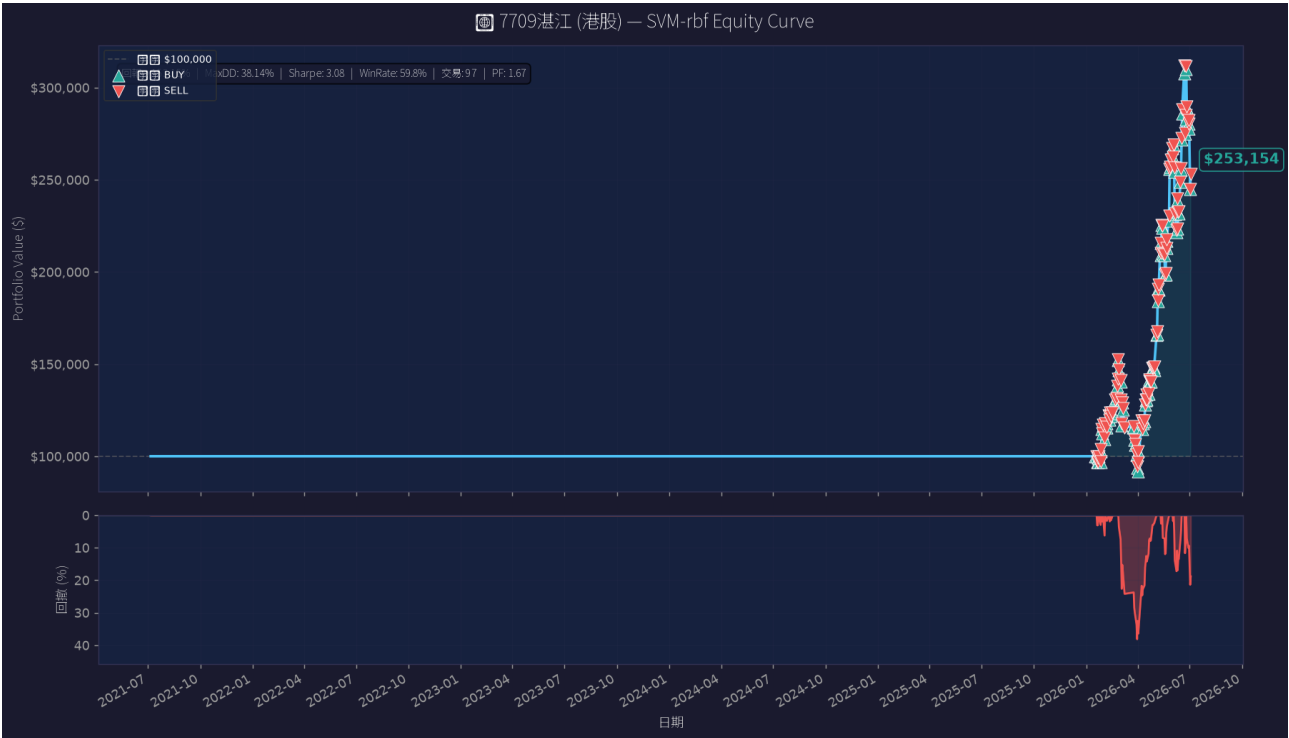
準確率	AccStd	Baseline	總回報	MaxDD	Sharpe	Sortino	Calmar	WinRate	ProfitF	AvgWin	AvgLoss	交易數
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60.00%	12.33%	61.82%	+153.15%	38.14%	3.08	5.59	4.02	59.8%	1.67	9.44%	-9.13%	97
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Fold 準確率: 45.5%, 63.6%, 45.5%, 54.5%, 45.5%, 72.7%, 81.8%, 72.7%, 54.5%, 63.6% | 參數: {"C": "0.1", "gamma": "scale", "kernel": "rbf", "max_iter": "5000", "probability": "True"}

基本面:

☒ 特徵重要性 (RF): upper_shadow:0.065 | ret_1d:0.049 | bb_width:0.037 | obv_ratio:0.037 | ret_10d:0.034 | lower_shadow:0.030 | ema12:0.029 | ema_diff:0.029



MU (US.MU) [美股]

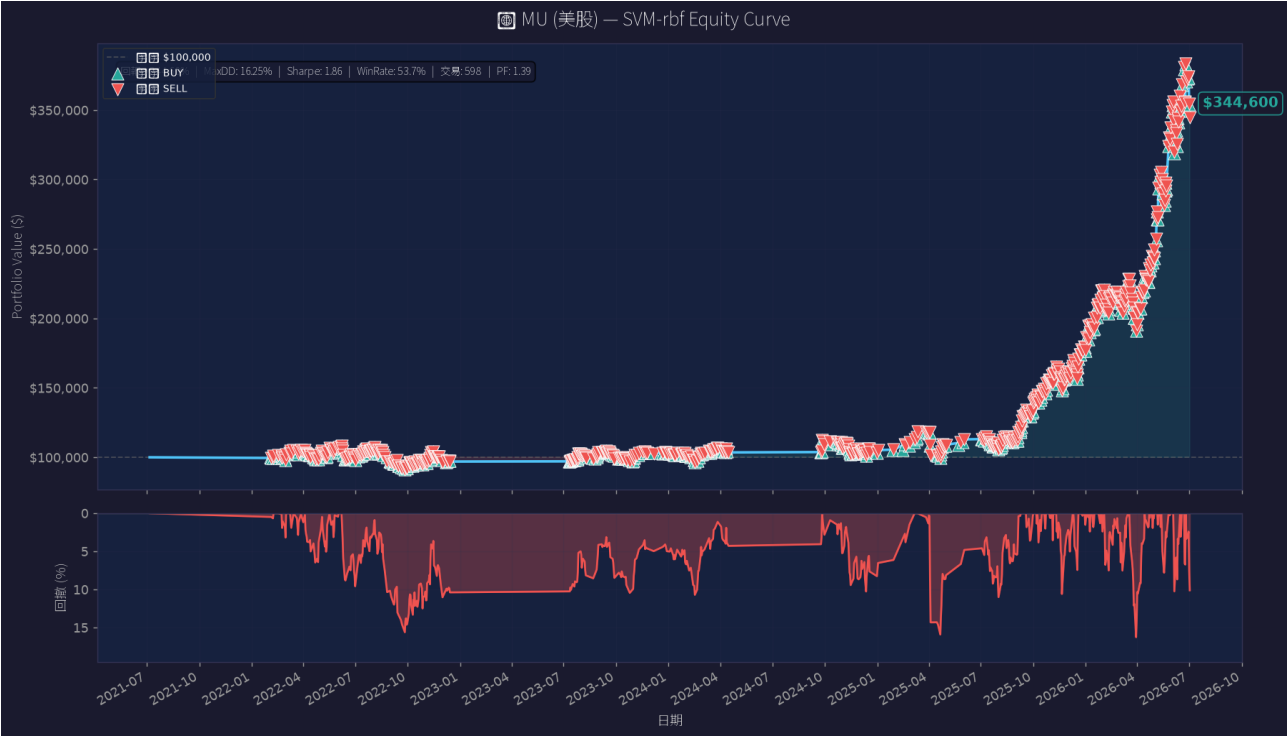
☒ 2021-08-10 ~ 2026-07-01 | 1228 bars | 41因子 | 基本面評分: 67.0

準確率	AccStd	Baseline	總回報	MaxDD	Sharpe	Sortino	Calmar	WinRate	ProfitF	AvgWin	AvgLoss	交易數
52.43%	3.75%	54.05%	+244.60%	16.25%	1.86	2.97	15.05	53.7%	1.39	3.02%	-2.52%	598

Fold 準確率: 49.5%, 49.5%, 56.8%, 51.4%, 46.8%, 50.5%, 54.1%, 51.4%, 60.4%, 54.1% | 參數: {"C": "10", "gamma": "scale", "kernel": "rbf", "max_iter": "5000", "probability": "True"}

基本面: pe_ttm=22.051 | pb=10.938 | roe=66.6381 | market_cap=1101790782389.56 | log_market_cap=27.727957955993926

☒ 特徵重要性 (RF): vol_ratio_5:0.038 | lower_shadow:0.035 | upper_shadow:0.034 | range_pct:0.031 | body_pct:0.031 | obv:0.029 | obv_ratio:0.029 | vol_ma5:0.028



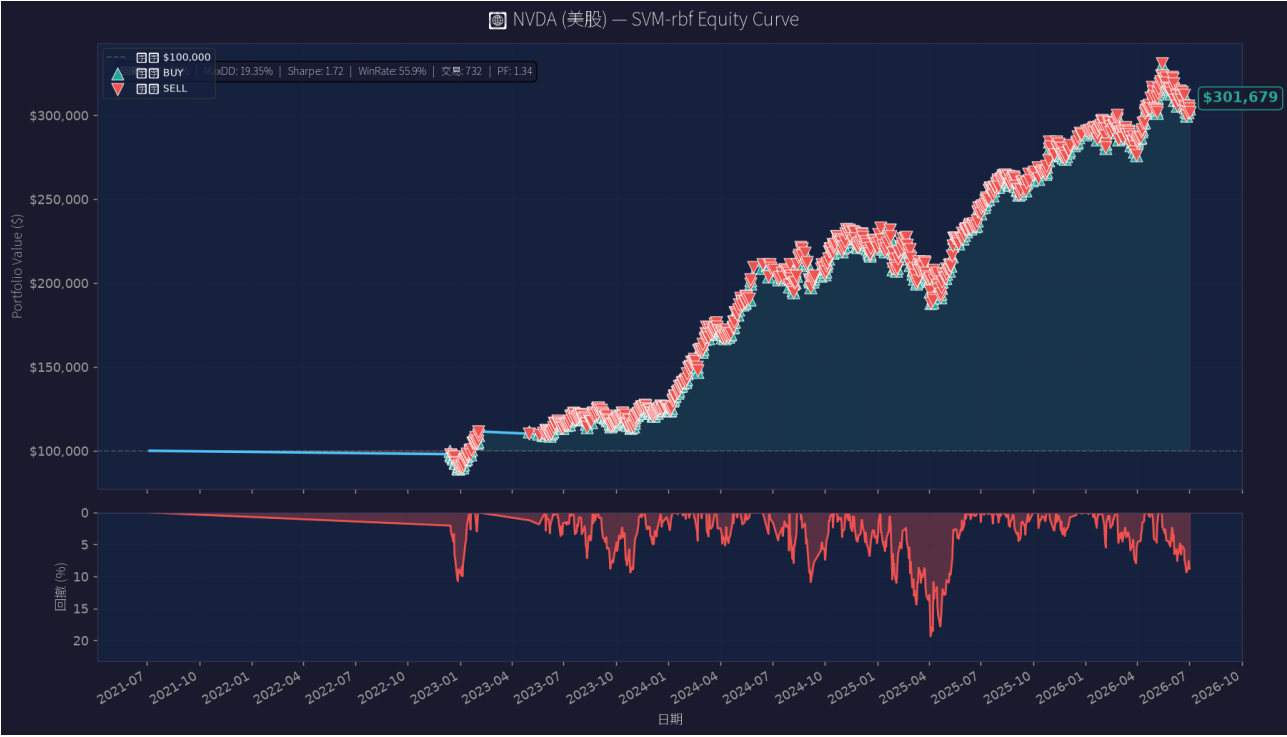
NVDA (US.NVDA) [美股]

☑ 2021-08-10 ~ 2026-07-01 | 1228 bars | 41因子 | 基本面评分: —

準確率	AccStd	Baseline	總回報	MaxDD	Sharpe	Sortino	Calmar	WinRate	ProfitF	AvgWin	AvgLoss	交易數
54.14%	4.62%	54.14%	+201.68%	19.35%	1.72	2.61	10.42	55.9%	1.34	2.27%	-2.15%	732

Fold 準確率: 52.3%, 46.8%, 47.7%, 52.3%, 60.4%, 58.6%, 55.0%, 56.8%, 60.4%, 51.4% | 參數: {"C": "0.1", "gamma": "scale", "kernel": "rbf", "max_iter": "5000", "probability": "True"}

☑ 特徵重要性 (RF): upper_shadow:0.038 | volatility_20d:0.036 | vol_ma5:0.033 | lower_shadow:0.032 | stoch_k:0.032 | range_pct:0.031 | vol_ratio_20:0.031 | bb_position:0.031



GOOGL (US.GOOG) [美股]

☑ 2021-08-10 ~ 2026-07-01 | 1228 bars | 41因子 | 基本面评分: 57.0

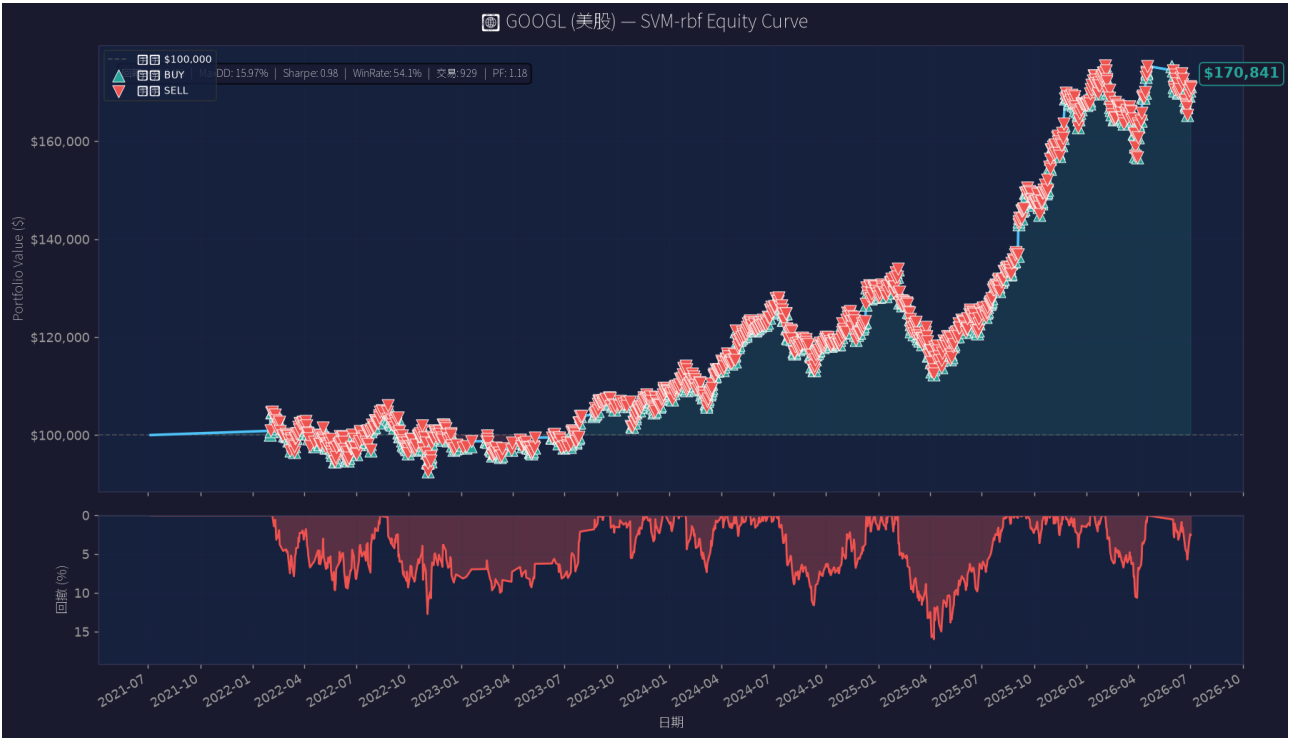
準確率	AccStd	Baseline	總回報	MaxDD	Sharpe	Sortino	Calmar	WinRate	ProfitF	AvgWin	AvgLoss	交易數
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54.14%	3.98%	55.32%	+70.84%	15.97%	0.98	1.45	4.43	54.1%	1.18	1.49%	-1.49%	929
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Fold 準確率: 52.3%, 51.4%, 49.5%, 57.7%, 58.6%, 59.5%, 54.1%, 54.1%, 57.7%, 46.8% | 參數: {"C": "10", "gamma": "scale", "kernel": "rbf", "max_iter": "5000", "probability": "True"}

基本面: pe_ttm=27.453 | pb=9.108 | roe=38.884800000000006 | market_cap=4360669560000.0 | log_market_cap=29.103646730306348

☒ 特徵重要性 (RF): vol_ratio_20:0.044 | stoch_k:0.037 | body_pct:0.035 | vol_ma5:0.034 | ret_20d:0.033 | ret_1d:0.033 | upper_shadow:0.032 | range_pct:0.032



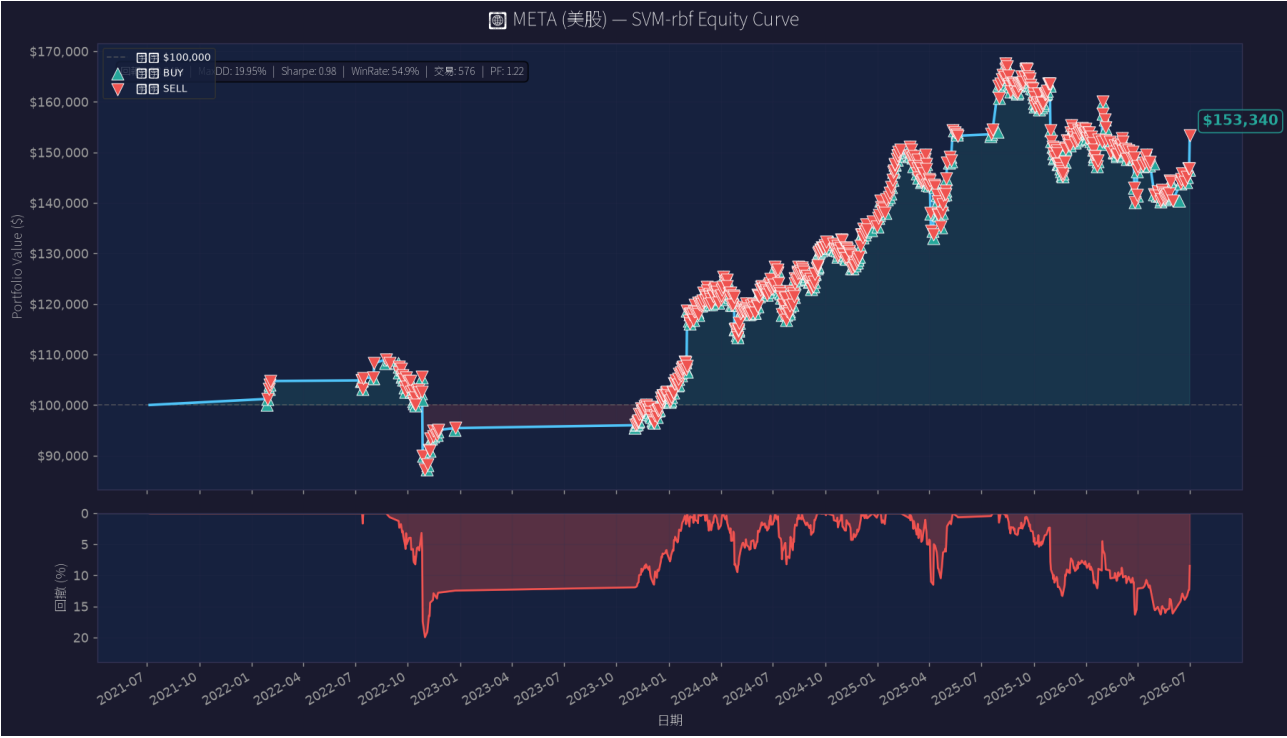
META (US.META) [美股]

☒ 2021-08-10 ~ 2026-07-01 | 1228 bars | 41因子 | 基本面評分: —

準確率	AccStd	Baseline	總回報	MaxDD	Sharpe	Sortino	Calmar	WinRate	ProfitF	AvgWin	AvgLoss	交易數
53.24%	3.87%	52.88%	+53.34%	19.95%	0.98	1.22	2.67	54.9%	1.22	1.71%	-1.72%	576

Fold 準確率: 55.9%, 55.0%, 47.7%, 47.7%, 55.9%, 56.8%, 56.8%, 50.5%, 48.6%, 57.7% | 參數: {"C": "1", "gamma": "scale", "kernel": "rbf", "max_iter": "5000", "probability": "True"}

☒ 特徵重要性 (RF): lower_shadow:0.044 | rsi_7:0.043 | volatility_5d:0.041 | upper_shadow:0.038 | stoch_k:0.034 | ret_1d:0.034 | vol_ratio_20:0.033 | macd_signal:0.028



MSFT (US.MSFT) [美股]

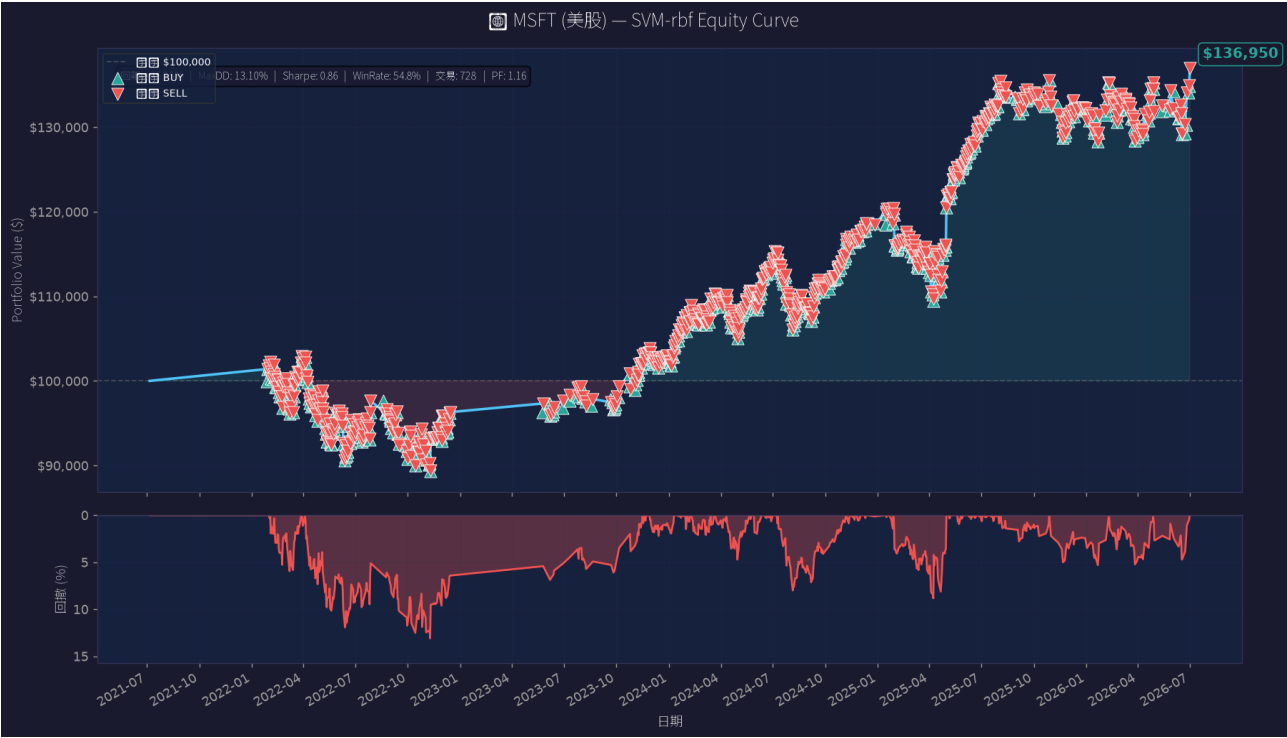
☑ 2021-08-10 ~ 2026-07-01 | 1228 bars | 41因子 | 基本面评分: 67.0

準確率	AccStd	Baseline	總回報	MaxDD	Sharpe	Sortino	Calmar	WinRate	ProfitF	AvgWin	AvgLoss	交易數
54.86%	4.16%	53.96%	+36.95%	13.10%	0.86	1.25	2.82	54.8%	1.16	1.23%	-1.29%	728

Fold 準確率: 48.6%, 54.1%, 51.4%, 51.4%, 52.3%, 57.7%, 64.0%, 56.8%, 57.7%, 55.0% | 參數: {"C": "0.1", "gamma": "scale", "kernel": "rbf", "max_iter": "5000", "probability": "True"}

基本面: pe_ttm=23.257 | pb=7.0 | roe=34.0142 | market_cap=2900729467564.96 | log_market_cap=28.6959833618296

☑ 特徵重要性 (RF): upper_shadow:0.041 | volatility_20d:0.034 | ret_1d:0.032 | vol_ratio_5:0.031 | obv_ratio:0.031 | ret_5d:0.030 | range_pct:0.029 | body_pct:0.028



⚠️ 免責聲明: 過往回測表現不代表未來結果。模型存在過擬合風險。50% 倉位、未計交易成本 (手續費/滑點)。基本面數據為最新 snapshot 並非歷史時序, pre-filter 為啟發式規則。